

Theoretical Computer Science Exam, September 13th, 2018

The exam consists of **4 exercises**. Available time: 1 hr 45 min.; you may write your answers in Italian, if you wish.

All exercises must be addressed for the written exam to be considered sufficient.

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I have passed the midterm exam and I intend to skip the first part (Ex.1-2) (select one): **YES NO**

Exercise 1 (8 pts)

Consider the following grammar G :

$S \rightarrow A B$
 $A \rightarrow a c C$
 $B \rightarrow C c b$
 $C \rightarrow c C \mid C c \mid c$
 $C C \rightarrow D$
 $D \rightarrow d D \mid d$
 $A B \rightarrow c C c$

- Identify the (smallest) language family of $L(G)$; provide a precise and concise description of $L(G)$.
- Design an automaton, of the least powerful type, that accepts $L(G)$.

Exercise 2 (8 pts)

Specify, by means of suitable pre- and post-conditions, a subprogram **gcd_lcm**(a, b, c, out) where a, b, c are input numeric parameters that may be assumed to have positive integer values such that $a < b < c$, and out is a Boolean output parameter. After execution of the subprogram, the output parameter out is true if and only if one of the following conditions holds.

- a is the greatest common divisor of b and c
- c is the least common multiple of a and b

Exercise 3 (8 pts)

Consider the following set

$$S = \{ x \mid |\{ y \mid f_x(y) \neq \perp \}| \leq 1 \}$$

i.e., the set of TM indices x such that the computable function f_x is defined for no more than one value. Answer the following questions, providing suitable justifications for your reply.

1. Is set S decidable?
2. Is set S semidecidable?
3. Is set $\neg S$ (the complement of S) semidecidable?

Exercise 4 (9 pts)

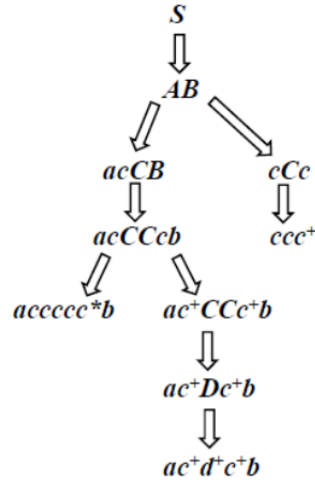
Consider the language, over the alphabet $\{0, 1\}$ that includes strings of the type xy , where $|x| = 5$ and x encodes in binary the position of a character, inside string y , which must be equal to 1.

Describe, in a clear, complete and unambiguous way, a single tape Turing machine and a RAM machine that accept the above language; evaluate the space and time complexity using all the available cost criteria.

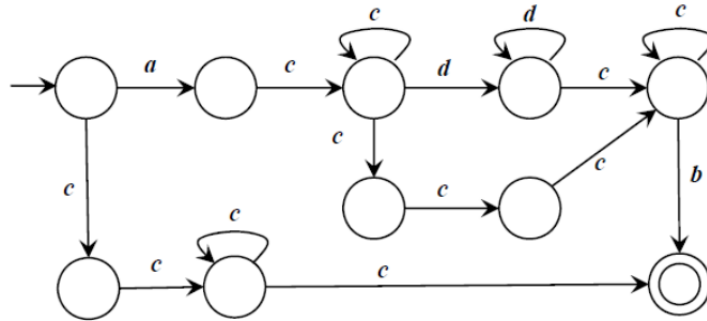
Solutions

Exercise 1

- a) Here below is reported a tree-like representation of all possible derivations. $L(G)$ is the regular language $\{accccc*b\} \cup \{ac^+d^+c^+b\} \cup \{ccc^+\}$.



- b) A (possibly nondeterministic) finite state automaton suffices.



Exercise 2

pre: $a > 0 \wedge b > 0 \wedge c > 0 \wedge a < b \wedge b < c$

post: out $\leftrightarrow$

$$\exists h1 \ h2 \ (b = a \cdot h1 \wedge c = a \cdot h2 \wedge \neg k \ k1 \ k2 \ (k > a \wedge b = k \cdot k1 \wedge c = k \cdot k2)) \ \vee$$

$$\exists h1 \ h2 \ (c = a \cdot h1 \wedge c = b \cdot h2 \wedge \neg k \ k1 \ k2 \ (k < c \wedge c = a \cdot k1 \wedge c = b \cdot k2))$$

Exercise 3

1. Is set S decidable? No, the Rice theorem applies.
2. Is set S semidecidable? It is **not** semidecidable by adopting the customary dovetailing simulation technique. Instead, the complement set, $\neg S = \{x \mid |\{y \mid f_x(y) \neq \perp\}| > 1\}$ is semidecidable, based on the dovetailing simulation technique (if $f_x(y)$ is defined for at least two values of y , then one can find them by means of dovetailing simulation). Therefore S cannot be semidecidable, otherwise S (and also $\neg S$) would be decidable.
3. Is set $\neg S$ (the complement of S) semidecidable? Yes, see answer to the previous point.

Exercise 4

The length of string x is fixed, hence the largest possible denoted position is 11111 in binary, that is, 31 in decimal. Hence the correct strings can be identified by considering only the prefix of length $5+31$; and the number of such prefixes is finite, therefore the language can be accepted by a finite state automaton, which terminates after 36 steps at most. Thus the TM and the RAM can work in the same way, simulating a finite state automaton, with the following complexity figures.

Single tape TM: $T(n) = \Theta(1)$, $S(n) = \Theta(n)$ (the latter because the input string is on the tape)

RAM, with both constant and logarithmic cost criteria: $T(n) = \Theta(1)$, $S(n) = \Theta(1)$.